

The storage capacity of the Ising perceptron: verification of the outstanding numerical conditions

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July 2026

Abstract

Krauth and Mézard (1989) predicted that the storage capacity of the Ising perceptron is $\alpha_\star N + o(N)$ for an explicit constant $\alpha_\star \approx 0.833$. Ding and Sun proved the matching lower bound, and Huang recently proved the matching upper bound; each result is conditional on a numerical hypothesis about an explicit low-dimensional variational problem, stated by those authors and checked by them only in floating point. Huang’s own rigorous interval arithmetic verified every numerical constant in his upper bound *except* the main Condition 1.3—the nonpositivity, over the whole plane, of a two-variable rate function whose supremum is not attained. That unbounded, degenerate optimization is the reason the capacity upper bound remained conditional.

We verify the condition in full. Reparametrizing Huang’s functional into the moment coordinates of the pair $(X, \tanh X)$ compresses the plane onto a *compact* convex body and turns the entropy term into a per-cell linear function by convex duality; certified sweeps then establish nonpositivity on the whole body away from a star-shaped region around the single degenerate maximizer. On the star itself no box enclosure can work—the tilted covariance $\nabla^2 \Phi$ is nearly singular and its inverse loses an essential cancellation—so we certify concavity of a majorant along rays from the maximizer, bounding the entropy Hessian by the explicit matrix $\mathbb{E}[f f^\top \max_\Lambda \operatorname{sech}^2(\lambda \cdot f)]$ over sublevel-set localizations of the dual that are themselves certified by one-dimensional sweeps; nothing is ever inverted. The same machinery, in one variable, verifies the Ding–Sun Condition 1.2 including the degenerate zero, and we re-establish the parameter rectangle $(\alpha_\star, q_\star, \psi_\star)$ on which both papers rest. Together these make the Krauth–Mézard prediction for the Ising perceptron capacity unconditional.

1 Introduction

Fix $\kappa = 0$ and let $g^1, g^2, \dots$ be independent standard Gaussian vectors in $\mathbb{R}^N$. The Ising perceptron is the random subset of the discrete cube

$$S_N^M = \{x \in \{-1, +1\}^N : \langle g^a, x \rangle \geq 0 \text{ for all } 1 \leq a \leq M\},$$

and its capacity is M_N/N , where M_N is the largest M with $S_N^M \neq \emptyset$. Krauth and Mézard [1] analyzed this model with the replica method and predicted that M_N/N converges in probability to an explicit constant $\alpha_\star \approx 0.83307860$, defined below. The prediction has been one of the standard test problems for making the cavity method rigorous ever since.

Rigorous progress converged on the constant from both sides. Ding and Sun [2] proved $\liminf_N \mathbb{P}(M_N/N \geq \alpha) > 0$ for every $\alpha < \alpha_*$; combined with the sharp threshold sequence of Xu [3] and Nakajima and Sun [4] (see also [6]), this gives the lower bound with high probability. Huang [5] proved $\mathbb{P}(M_N/N \geq \alpha) \rightarrow 0$ for every $\alpha > \alpha_*$. Both results are conditional. The Ding–Sun theorem assumes their Condition 1.2: a certain function $\lambda \mapsto \mathcal{S}_*(\lambda)$, arising as the exponential rate of a restricted second moment, is negative except at its two degenerate zeros $\lambda \in \{0, 1\}$. Huang’s theorem assumes his Condition 1.3: a two-variable function $(\lambda_1, \lambda_2) \mapsto \mathcal{S}_*(\lambda_1, \lambda_2)$, the rate function of a planted first moment, is nonpositive on all of $\mathbb{R}^2$, vanishing at the single point $(1, 0)$. Both papers verified their conditions numerically, in ordinary floating-point arithmetic, and stated them as hypotheses. Huang’s second arXiv version added rigorous interval arithmetic for every numerical constant in his proof except Condition 1.3 itself; Ding and Sun wrote that they intended to supply a rigorous verification in a later version of their paper, which did not appear.

This paper closes the remaining gap: we verify both conditions, and re-derive the parameter rectangle

$$\alpha_* \in [0.833078599, 0.833078600], \quad q_* \in [0.56394907949, 0.56394908030], \quad \psi_* \in [2.5763513100, 2.5763513200]$$

of [2, Proposition 1.3], on which every numerical statement in both papers (and in this one) rests, in rigorous interval arithmetic. Consequently:

Theorem 1.1. *For the Ising perceptron at $\kappa = 0$ with Gaussian disorder, $M_N/N \rightarrow \alpha_*$ in probability. The same holds for i.i.d. subgaussian disorder with mean zero and unit variance, in particular for ± 1 entries, by the universality theorem of [4].*

The verification is computer-assisted in the sense that finitely many inequalities between explicitly defined real numbers are established by interval arithmetic (we use the ball arithmetic of Arb via python-flint, with certified adaptive quadrature for all integrals). It is not, however, a matter of running an integrator over the stated conditions. Three genuinely mathematical obstructions have to be removed first, and most of this paper is devoted to them.

First, Huang’s condition is a statement over the whole plane, and the supremum at infinity is not attained: $\mathcal{S}_*$ tends to zero along the degenerate directions only polynomially. Reparametrizing by the moments of the pair $(X, \tanh X)$, $X \sim N(0, \psi_*)$ (§5), maps the plane onto a bounded convex body and removes the tail entirely, while convex duality turns the entropy term into a per-cell linear function; the bulk of the body is then certified by an adaptive sweep (§6).

Second, both variational functions attain the value zero at degenerate maximizers, where no direct interval evaluation can certify a sign, and near Huang’s maximizer no box enclosure of any Hessian of the problem can work either: the tilted covariance $\nabla^2 \Phi$ is nearly singular there, and its inverse—which the curvature of the entropy term is—loses an essential cancellation in interval arithmetic. We certify concavity of a majorant along rays from the maximizer instead, bounding the entropy Hessian by an explicit matrix over sublevel-set localizations of the dual variable; nothing is ever inverted (§6). The pinned value and gradient at the center are exact identities of the fixed point and require no numerics. For the Ding–Sun condition the degenerate points are handled by their own derivative and second-derivative analysis and a corrected near- $\lambda = 1$ chain, whose numerical inputs we certify (§4, §7).

Third, the interval evaluations themselves have to be organized so that widths do not defeat the margins, which are as small as 10^{-3} in the interior of the Ding–Sun grid and near the star boundary of the Huang sweep. Every delicate cell is evaluated in exact mean-value form—value at the center plus an enclosed gradient times the radius—so that the near-cancellations the margins depend on survive enclosure; the exact identities $\mathcal{H}'(\lambda) = -(1-q_\star) \log A(\lambda)/2$ and the explicit derivative integrals supply the gradients, and parameter-ball evaluation subsumes the manual corner analyses mechanically.

The verification programs, the logs of the certified runs, and instructions to reproduce every number in this paper are available at [7]. All decimal constants enter the programs as exact rationals; all comparisons are between balls whose correctness is guaranteed by Arb. We follow the numbering of [2] and [5] for their statements.

2 The statements being verified

We use the normalizations of [2, 5]. With φ, Ψ the standard Gaussian density and upper tail, and $\mathcal{E} = \varphi/\Psi$, set for $q \in [0, 1)$

$$F_{1-q}(x) = \frac{\mathcal{E}}{\sqrt{1-q}} \left(\frac{-x}{\sqrt{1-q}} \right), \quad P(\psi) = \mathbb{E} \tanh^2(\sqrt{\psi}Z), \quad R_\alpha(q) = \alpha \mathbb{E} F_{1-q}(\sqrt{q}Z)^2,$$

and the Gardner free energy $\mathcal{G}(\alpha, q, \psi)$ of [2, eq. (1.6)]. The threshold $\alpha_\star$ is the root of $\mathcal{G}$ along the fixed point $q = P(\psi)$, $\psi = R_\alpha(q)$; Ding–Sun Proposition 1.3 makes this precise and pins the rectangle of §3. Huang’s Condition 1.3 is $\mathcal{S}_\star \leq 0$, the object of §5–6; Ding–Sun’s Condition 1.2 is $\mathcal{S}_\star(\lambda) < 0$ for $\lambda \notin \{0, 1\}$, the object of §4. §7 records which theorem consumes which verified fact.

3 The parameter rectangle

Block 1 (`block1_gardner.py`) re-proves Ding–Sun Proposition 1.3 in interval arithmetic: thirteen certified inequalities establishing (i) that R_α maps the stated q -intervals into the stated ψ -intervals; (ii) the Almeida–Thouless contraction $\sup_G dP(R_\alpha)/dq \leq 0.96 < 1$, via $P' \leq 0.08$ and $dR/dq \leq 12$; (iii) the fixed-point sign changes $P(R(q, \alpha)) - q$ at the four corners; and (iv) $\mathcal{G}_\star(\alpha_{\text{ub}}) < 0 < \mathcal{G}_\star(\alpha_{\text{lb}})$. Each is a one-dimensional certified Gaussian integral (adaptive `acb.integral` plus an explicit tail bound), evaluated with α, q, ψ as balls covering their whole ranges, which subsumes Ding–Sun’s manual corner analysis. This establishes $\alpha_\star \in [0.833078599, 0.833078600]$ and the accompanying $q_\star, \psi_\star$ intervals rigorously.

4 The Ding–Sun condition

Ding–Sun’s Condition 1.2, $\mathcal{S}_\star(\lambda) = \mathcal{H}(\lambda) + \mathcal{P}(\lambda) + \mathcal{A}(\lambda) < 0$ for $\lambda \notin \{0, 1\}$, is one-dimensional and mirrors Huang’s: a value sweep on the bulk and a local second-derivative analysis at the two degenerate zeros $\lambda = 0, \lambda = 1$. Since $\mathcal{A} \leq 0$, it suffices to bound $\mathcal{H} + \mathcal{P}$ on the positive branch and, on the negative branch, the tilted majorant $\mathcal{H} + \mathcal{Q}$ (Ding–Sun’s $s = 0.2$ shift). We build the certified evaluators (`dsfun.py`): $\mathcal{H}(\lambda) = \mathfrak{H}(A(\lambda))$ through the pair-entropy Γ and the map $A = \exp(2 \operatorname{atanh} \lambda)$, with the well-conditioned $D_H(A) = (1 - m^2)(1 - 2/(\Delta + 1))$ that

removes the interval blow-up of the raw $(A^2 - 1)/(\Delta + 1)^2$ form; and $\mathcal{P}, \mathcal{Q}$ through the double integral $I_s(\lambda)$, evaluated by nested certified quadrature with mean-value corrections. The bulk cells verify. Near $\lambda = 1$ we follow Ding–Sun’s own chain (their Lemma 8.2, Corollary 8.3, Propositions 8.4 and 8.1) with corrected constants: the two integrals behind their Lemma 8.2 evaluate to 2.678... and 3.162..., not the printed 1.78 and 4.3 (the printed consequence $1 - \ell(A) \leq 1.83/A$ is already inconsistent with their verified $1 - \ell(100) > 0.025$), and the double integral of their Proposition 8.4 is $-0.4447...$ rather than -0.45 . With the certified values ($1 - \ell(A) \leq 2.711/A$ for $A \geq 100$, corollary constant 0.9987, closing coefficient 1.2837) the same proof gives $\mathcal{H} + \mathcal{P} < \mathcal{H}(1) + \mathcal{P}(1)$ on $[0.982, 1)$, and the part (a) grid reaches $\lambda(0.99) = 0.98665... > 0.982$, so the covering of $[\lambda_{\min}, 1)$ is unchanged. The neighborhood of $\lambda = 0$ is handled by the derivative and second-derivative certificates described in §7.

5 Huang’s condition in moment coordinates

Huang’s Condition 1.3 asks that

$$\mathcal{S}_\star(\lambda_1, \lambda_2) = \inf_{s \geq 0} \left\{ \frac{1}{2} s^2 \psi_0 + \text{ent}(\Lambda_{\lambda_1, \lambda_2}) + \alpha_\star \mathbb{E} \log \Psi(V_s) \right\} \leq 0 \quad \text{for all } (\lambda_1, \lambda_2) \in \mathbb{R}^2,$$

where $\Lambda_{\lambda_1, \lambda_2}(x) = \tanh(\lambda_1 x + \lambda_2 \tanh x)$, $X \sim N(0, \psi_0)$, $M = \tanh X$, and the constraint term depends on the profile only through the two moments

$$a_1 = \mathbb{E}[X \Lambda], \quad a_2 = \mathbb{E}[M \Lambda].$$

The plane $\mathbb{R}^2$ of (λ_1, λ_2) is unbounded, and the supremum is not attained: along the degenerate directions $\mathcal{S}_\star \rightarrow 0^-$ only polynomially. Verifying a nonpositivity statement over an unbounded domain by interval arithmetic is the obstruction that stopped a direct check.

We remove it by changing coordinates. The profile $\Lambda_{\lambda_1, \lambda_2}$ is exactly the maximum-entropy profile for its own moments (this is the Lagrange condition that produced the two-parameter family in the first place), so

$$\text{ent}(\Lambda_{\lambda_1, \lambda_2}) = H(a_1, a_2) := \max \left\{ \mathbb{E} \text{ent}_2\left(\frac{1+\Lambda}{2}\right) : \mathbb{E}[X \Lambda] = a_1, \mathbb{E}[M \Lambda] = a_2, |\Lambda| \leq 1 \right\},$$

and therefore

$$\sup_{\lambda_1, \lambda_2} \mathcal{S}_\star(\lambda_1, \lambda_2) = \sup_{(a_1, a_2) \in K} [H(a_1, a_2) + G(a_1, a_2)], \quad G(a_1, a_2) = \inf_{s \geq 0} \left\{ \frac{1}{2} s^2 \psi_0 + \alpha_\star \mathbb{E} \log \Psi(V_s) \right\}. \quad (1)$$

The key point is that the new domain

$$K = \{(\mathbb{E}[X \Lambda], \mathbb{E}[M \Lambda]) : |\Lambda| \leq 1\}$$

is the *moment body* of the pair (X, M) , a compact convex set with support function $h(u, v) = \sup_{|\Lambda| \leq 1} \mathbb{E}[\Lambda(uX + vM)] = \mathbb{E}[uX + vM]$. It is contained in the rectangle $|a_1| \leq \mathbb{E}|X|$, $|a_2| \leq \mathbb{E}|M|$. The unbounded tail of the (λ_1, λ_2) plane is compressed onto ∂K , where the profiles are saturated ($|\Lambda| = 1$) and hence $H \upharpoonright_{\partial K} = 0$.

Two consequences make (1) checkable. First, the constraint term G depends on (a_1, a_2) explicitly, through $V_s = (-\frac{a_2}{q_0} \sqrt{q_0} Z - \frac{a_1}{\psi_0} \mathbf{N})/D + s \mathbf{N}$ with Z standard Gaussian, $\mathbf{N} =$

$\mathcal{E}(-\gamma Z)/\sqrt{1-q_0}$ for $\gamma = \sqrt{q_0/(1-q_0)}$, and $D = \sqrt{1-a_2^2/q_0}$, a one-dimensional Gaussian integral; and $G \leq \frac{1}{2}s^2\psi_0 + \alpha_\star \mathbb{E} \log \Psi(V_s)$ for *any* fixed s , so a per-cell choice of s gives a certified upper bound with no inner optimization. Second, convex duality gives, for *every* dual point (b_1, b_2) ,

$$H(a_1, a_2) \leq \Phi(b_1, b_2) - b_1 a_1 - b_2 a_2, \quad \Phi(b_1, b_2) = \mathbb{E} \log 2 \cosh(b_1 X + b_2 M), \quad (2)$$

with equality when (b_1, b_2) is the dual of (a_1, a_2) . Choosing (b_1, b_2) per cell (found by a Newton solve on the convex dual objective, then used as a fixed rational) turns the entropy into an explicit linear function of (a_1, a_2) , tight on the cell.

Lemma 5.1 (Moment reduction). *With H and G as above, $\sup_{\lambda_1, \lambda_2} \mathcal{S}_\star = \sup_{(a_1, a_2) \in K} (H + G)$, and (2) holds for all (b_1, b_2) .*

Proof. Fix (λ_1, λ_2) and write $\Lambda = \Lambda_{\lambda_1, \lambda_2}$, $a_i = a_i(\lambda)$. The constraint term of $\mathcal{S}_\star$ depends on Λ only through (a_1, a_2) (it enters $\mathcal{S}_\star(\Lambda, s)$ only via $\mathbb{E}[M\Lambda] = a_2$ and $\mathbb{E}[X\Lambda] = a_1$, in the notation of §5), so $\mathcal{S}_\star(\lambda_1, \lambda_2) = \text{ent}(\Lambda) + G(a_1, a_2)$. It remains to show $\text{ent}(\Lambda) = H(a_1, a_2)$, i.e. that Λ maximizes the entropy among all profiles with moments (a_1, a_2) . This is the Lagrange condition that defines Λ : maximizing $\mathbb{E} \text{ent}_2(\frac{1+\Lambda}{2})$ over $|\Lambda| \leq 1$ subject to $\mathbb{E}[X\Lambda] = a_1$, $\mathbb{E}[M\Lambda] = a_2$ has, by pointwise optimization of the Lagrangian $\text{ent}_2(\frac{1+\Lambda}{2}) + (\lambda_1 X + \lambda_2 M)\Lambda$, the maximizer $\Lambda^* = \tanh(\lambda_1 X + \lambda_2 M) = \Lambda$. Hence $\text{ent}(\Lambda) = H(a_1, a_2)$ and $\mathcal{S}_\star(\lambda_1, \lambda_2) = (H + G)(a_1(\lambda), a_2(\lambda))$. Taking suprema and noting that $(a_1(\lambda), a_2(\lambda))$ ranges over a dense subset of K (every extreme point of K is a limit of such profiles) gives the first claim. For (2), strong duality for this concave program reads $H(a) = \min_b [\Phi(b) - b \cdot a]$, where the inner maximization of the Lagrangian over $|\Lambda| \leq 1$ gives $\max_\Lambda [\text{ent}_2(\frac{1+\Lambda}{2}) + \theta \Lambda] = \log 2 \cosh \theta$ at $\theta = b_1 X + b_2 M$; the stated inequality is the weak-duality direction, valid for every b . $\square$

Region II certifies, cell by cell, the inequality $\min_k [\Phi(b^k) - b^k \cdot a] + \frac{1}{2}(s^C)^2 \psi_0 + \alpha_\star T_C(a) < 0$ on the cell, where T_C encloses $\mathbb{E} \log \Psi(V_{s^C})$ over the cell by a mean-value rule in (a_1, a_2) . By Lemma 5.1 the first term bounds H and the rest bounds G from above on the cell, so the certified inequality gives $(H + G) < 0$ there. Cells whose closure misses K are dropped: a cell lies outside K once some direction (u, v) has $\min_{a \in \text{cell}} (ua_1 + va_2) > h(u, v)$, which is checked against a certified upper bound on $h(u, v) = \mathbb{E}[uX + vM]$ over a fan of directions.

6 The compact sweep and the degenerate point

Write $a^\star = (\psi_0(1-q_0), q_0)$ for the image of $(\lambda_1, \lambda_2) = (1, 0)$; it is the unique global maximizer, with $H(a^\star) + G(a^\star) = 0$. We verify (1) in two pieces.

Region II (the bulk). For every cell C of a bounded, adaptively refined grid on K away from the star region of Region I, we certify

$$[\Phi(b^C) - b_1^C a_1 - b_2^C a_2] + \frac{1}{2}(s^C)^2 \psi_0 + \alpha_\star T_C < 0 \quad \text{on } C,$$

where b^C, s^C are the (nonrigorous) per-cell dual and tilt and T_C is a mean-value enclosure of $\mathbb{E} \log \Psi(V_{s^C})$ over C . Cells lying outside K (detected by the support function h along a fan of directions) contribute nothing and are dropped; cells straddling ∂K are handled by pulling the dual inward, using that (2) holds for any fixed b . The sweep runs in two stages—all of K

minus a coarse box around $a^\star$, then that box minus the star, bisecting to side 10^{-3} along the star boundary—and a cell is skipped only if a conservative test places it entirely inside the star (maximal corner distance below the minimal star radius over the cell’s angular range). Both stages terminate with every leaf cell strictly negative.

Region I (the maximizer). Near the maximizer the value is genuinely 0 at $a^\star$, so no direct sign check can succeed there, and interval evaluation of any Hessian of the problem over a two-dimensional box fails for a structural reason: $\nabla^2\Phi(1,0)$ has condition number ≈ 82 (the pair $(X, \tanh X)$ is strongly correlated), so the dual image of an a -box is stretched by a factor ≈ 220 along one direction and every box enclosure of $\nabla^2 H = -[\nabla^2\Phi]^{-1}$ loses the cancellation it needs. We work instead along rays. For a unit direction v and a fixed slope $\dot{\sigma}$ set

$$\varphi_v(t) = H(a^\star + tv) + \frac{1}{2}s(t)^2\psi_0 + \alpha_\star T(a^\star + tv, s(t)), \quad s(t) = s_0 + t\dot{\sigma}, \quad s_0 = \sqrt{1 - q_0}.$$

Since G is a minimum over the tilt, $\varphi_v \geq \mathcal{S}_\star$ along the ray, while at the fixed point (Huang’s unconditional identities: the Gardner stationarity and the tilt equation) $\varphi_v(0) = 0$ and $\varphi'_v(0) = 0$. Hence certifying $\varphi''_v \leq 0$ on $[0, T(v)]$ proves $\mathcal{S}_\star \leq 0$ on the segment, and a fan of direction cells covering the circle proves it on the star region $\{a^\star + tv : 0 \leq t \leq T(v)\}$, which contains every cell the stage-2 sweep skips. The second derivative splits into a constraint part—certified fixed-grid integrals over the (thin) ray piece, in which no dual variable appears—and the entropy part $-v^\top[\nabla^2\Phi(\lambda(x))]^{-1}v$, which we bound WITHOUT tracking the ill-conditioned dual map: since $\nabla^2\Phi(\lambda) = \mathbb{E}[ff^\top \text{sech}^2(\lambda \cdot f)]$ with $f = (X, \tanh X)$, any localization $\lambda(x) \in \Lambda$ gives

$$\nabla^2\Phi(\lambda(x)) \preceq B_\Lambda := \mathbb{E}[ff^\top \max_{\lambda \in \Lambda} \text{sech}^2(\lambda \cdot f)], \quad \text{so} \quad v^\top \nabla^2 H v \leq -v^\top B_\Lambda^{-1} v,$$

and B_Λ is a single explicit one-dimensional integral (the inner maximum is a per- z corner selection). With $\Lambda = \mathbb{R}^2$ this is the global bound $B_0 = \mathbb{E}ff^\top$, which settles every direction outside a narrow window ($\approx \pm 0.095$ rad) around the weak eigenvector of $\nabla^2\mathcal{S}_\star(a^\star)$. Inside the window the localization Λ is supplied per radius band by a sublevel-set argument that never inverts anything: $\lambda(x)$ minimizes $\Phi(\lambda) - \lambda \cdot x$, so it lies in the convex set $\{\lambda : \Phi(\lambda) - \lambda \cdot x \leq \min_k [\Phi(\hat{\lambda}_k) - \hat{\lambda}_k \cdot x]\}$ for any finite anchor fan $\hat{\lambda}_k$, and a certified one-dimensional sweep of Φ over the boundary of a candidate box (in exact mean-value form, with the per-segment derivative $\nabla\Phi - x$ enclosed—it is genuinely small along the flat valley) verifies that this set stays inside the box for every x in the band; the test function is convex in x , so corner points of the band hull suffice. The boxes shrink quadratically-in-radius toward $(1,0)$, so $B_\Lambda \rightarrow \nabla^2\Phi(1,0)$ exactly where sharpness is needed. All certificates run over the parameter balls of the Ding–Sun rectangle, hence cover the true fixed point.

7 Assembly

The pieces combine as follows.

Upper bound. Huang’s Theorem 1.1 reduces the capacity upper bound at $\kappa = 0$ to four numerical conditions. His other three numerical conditions (the parameter rectangle, the AMP constants, and the local concavity constants) were verified by Huang in `python-flint` and, for the first, rest on Ding–Sun Proposition 1.3, which we re-verify in §3 (Block 1). His Condition 1.3 is the statement $\mathcal{S}_\star \leq 0$ on $\mathbb{R}^2$. In moment coordinates (§5) this is $\sup_K(H +$

$G) \leq 0$; the two sweeps of Region II (§6) certify $H + G < 0$ on K outside the certified star region around a^* (the first over K minus a coarse box, the second over that box minus the star, bisecting to side 10^{-3} along the star boundary), and Region I’s ray certificates give $H + G \leq 0$ on the star itself. Hence $\sup_K(H + G) = 0$, i.e. Condition 1.3 holds, and Huang’s upper bound $\lim_N \mathbb{P}(M_N/N \geq \alpha) = 0$ for $\alpha > \alpha_*$ becomes unconditional.

Lower bound. Ding–Sun’s main theorem gives, under their Condition 1.2, that $\liminf_N \mathbb{P}(M_N/N \geq \alpha) > 0$ for $\alpha < \alpha_*$; the sharp-threshold results of Xu and Nakajima–Sun upgrade this to high probability. Condition 1.2 is the one-dimensional statement $\mathcal{S}_*(\lambda) < 0$ for $\lambda \notin \{0, 1\}$, and the covering of $[\lambda_{\min}, 1]$ follows their own decomposition: the value grids on $[0.2, 0.98]$ and $[\lambda_{\min}, -0.125]$ (part (a), our Block 3a), the derivative signs $\text{PG}' < 0$ on $[0.05, 0.2]$ and $\text{PG}' > 0$ on $[-0.125, -0.03]$ (part (b)), the second derivative $\text{PG}'' < 0$ on $[-0.03, 0.05]$ around the degenerate zero $\lambda = 0$ where $\text{PG}(0) = \text{PG}'(0) = 0$ (part (c)), and the near-one analysis on $[0.98, 1)$ (Block 2). Parts (b) and (c) use the exact derivative identities $\mathcal{H}' = -(1 - q_*) \log A/2$, $\mathcal{H}'' = -(1 - q_*)/(2A \ell'(A))$ and certified two-dimensional integrals for I' and I'' (with closed-form Gaussian-moment bounds for the mass outside the integration box); the interval endpoints are pinned by the certified monotone map $\lambda = \ell(A(\tau))$, exactly as in part (a). On the negative interval of part (b) the margins are uniformly thin (about 2×10^{-3}) while a ball-parameter evaluation of I' wraps the cell width by two orders of magnitude, so the cells there are certified in mean-value form: I' at the exact cell center, plus a bound on $|\text{PG}''|$ times the cell radius, the latter assembled from certified point values of I'' on a fine grid together with a closed-form bound on $|I'''|$ (a degree-four Gaussian-moment estimate) that controls the values between grid points.

Disorder. Krauth and Mézard’s model has ± 1 (Bernoulli) disorder; we work with Gaussian disorder throughout. The two agree because the capacity has a universal sharp threshold sequence over subgaussian disorder (Nakajima–Sun), which includes both.

What is machine-checked. Every statement labelled a Claim is a finite list of ball inequalities between explicitly defined real numbers, established by Arb; the analytic reductions around them (the moment-body reduction, convex duality, the ray majorants and their pinned identities, the localization and B_Λ arguments, the reductions to certified quadrature) are proved on paper. The nonrigorous companion (`huang_np.py`) only selects per-cell duals, tilts, anchors, and candidate boxes, which enter the certificates as fixed rationals; no certified inequality depends on it.

Inventory. The final runs comprise: the parameter rectangle, 13 checks; Huang Region II, 873 leaf cells (stage 1, the moment body minus a coarse box) and 1337 leaf cells (stage 2, the box minus the star), all strictly negative; Huang Region I, 836 band certificates covering the star with zero failures; Ding–Sun part (a), 118 grid cells; the corrected near-one block, 10 checks; and the middle-interval derivative and second-derivative certificates. Logs and JSON summaries for every run are included with the source.

Remark 7.1 (Soundness audit). *Two implementation errors were caught during this work by independent cross-checks and corrected before the final runs; we record them because they illustrate what interval arithmetic does and does not protect against. First, an early version of the fixed-grid quadrature multiplied the derivative enclosure of the mean-value remainder by the signed first moment $\int_{\text{cell}} (z - m) \varphi dz \approx 0$; since the enclosed derivative varies over the cell while $(z - m)$ changes sign, the remainder does not factor through the signed moment, and the resulting balls were systematically too tight—detected because a 30-digit quadrature*

of $\nabla\Phi(1,0)$ fell outside them. The corrected rule handles the positive and negative parts of $(z - m)$ separately. Second, a sign error in ∂_s^2 of the tilted constraint term (+ for $-$) survived until a finite-difference cross-check of every ingredient of φ_v'' against an independent floating-point implementation. Neither error is visible to the interval machinery itself: Arb guarantees the arithmetic, not the formulas. All certificates reported here postdate both corrections.

Funding and declarations

The author acknowledges support from the European Research Council under the European Union’s Horizon 2022 research and innovation programme, the Simons Foundation, Heights Labs, and Israel Science Foundation grants 2258/19 and 4101/25.

Most of the work reported here was done by Fable 5 (Anthropic), an AI system: the mathematical theory (the moment-coordinate reduction, the ray majorants and their pinned identities, the B_Λ bound and the sublevel-set localization, the corrected near-one chain), essentially all of the code, the soundness audits, and the drafting of this manuscript. The system operated in an autonomous goal mode: the author set the objective of resolving the outstanding conditions, and it worked toward that objective over extended unattended sessions, choosing the approach, implementing and running the certificates, detecting and repairing its own errors through independent cross-checks, and iterating until the verifications closed. The author supervised the runs, reviewed the mathematics, and accepts responsibility for the final manuscript. This is part of a broader research project of the author on the extent to which current AI systems can resolve open problems in mathematics.

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